Giridhari Pal

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Professional Summary

Backend Software Engineer with 3+ years of production experience designing, building, and scaling backend services and RESTful/gRPC APIs in Python and Rust. Proven expertise in microservices architecture, event-driven systems (Apache Kafka, MQTT), and relational/NoSQL databases (PostgreSQL, MongoDB, Redis, InfluxDB). Skilled in containerized deployments with Docker, cloud infrastructure (AWS, DigitalOcean), and building fault-tolerant, high-throughput data pipelines processing millions of records daily at sub-100ms latency.

Technical Skills

Languages: Python, Rust
Backend & API Development: FastAPI, Django, Flask, Actix-Web, Node.js, REST API, gRPC, Microservices
Architecture: Event-Driven Architecture, Distributed Systems, Asynchronous Processing, System Design
Databases: PostgreSQL, MongoDB, Redis, InfluxDB, DuckDB, SQLite
Messaging & Streaming: Apache Kafka, MQTT, Node-RED
Cloud & DevOps: AWS (EC2, S3), DigitalOcean, Utho Cloud, Docker, Linux (Ubuntu Server), Git
Frontend & Tools: React.js, TypeScript, Tauri, PyQt5, Grafana, Metabase, Power BI

Professional Experience

Junior Research Fellow — Backend & Data Systems Aug 2025 – Present
Dhirubhai Ambani University *Gujarat, India*

- Own end-to-end backend architecture for an Energy Management System, integrating ESP32-based IoT devices with backend services for reliable, continuous telemetry ingestion from distributed hardware.
- Design and deploy containerized microservices using Docker, FastAPI, gRPC, and Actix-Web with PostgreSQL, InfluxDB, and Redis to support scalable, fault-tolerant data processing.
- Architect real-time data pipelines using MQTT and Kafka to support high-throughput telemetry ingestion with minimal latency.
- Built a secure Actix-Web OTA update service handling firmware versioning, compatibility validation, and controlled rollout, improving device reliability.

Freelance Backend Engineer / Technical Consultant Oct 2024 – Aug 2025
Independent Consulting *Remote*

- Architected and deployed custom ERP and Leave Management systems using FastAPI, PostgreSQL, and React, with Docker-based deployment on cloud infrastructure.
- Designed REST API integrations connecting ZWCAD with FastAPI backends (via VBA and Tauri) to automate design-to-inventory workflows for manufacturing clients.
- Built backend services for ESP32-based embedded systems supporting sensor data ingestion and motor control logic.

SDE & DevOps Consultant / Intern Nov 2021 – Oct 2024
RC Labs — Battery Management System *Bangalore, India*

- Owned backend and data pipeline architecture integrating IoT telemetry across desktop, web, and cloud environments, processing millions of data points daily at sub-100ms latency.
- Built REST APIs connecting web applications to PostgreSQL and MongoDB with secure, encrypted communication.
- Implemented multithreaded and multiprocessing Python services, improving system throughput and fault tolerance.
- Standardized deployments using Docker and centralized Grafana dashboards for reliable remote monitoring across production environments.

Key Projects

ERP & Leave Management Systems — FastAPI, PostgreSQL, React, Docker

- Built production ERP platforms for manufacturing clients and a role-based Leave Management System with automated email notification workflows.

Inventory Management Desktop Application — Tauri, FastAPI, React, TailwindCSS

- Built a cross-platform inventory application integrating CAD drawing data via VBA automation for seamless design-to-inventory mapping.

Bidding Application — Python, PyQt5, PostgreSQL

- Developed a desktop application for real-time cricket bidding with relational data persistence and a responsive UI.

Education

Sri Krishna Institute of Technology (VTU) 2019 – 2023
B.E. in Information Science and Engineering (CGPA: 7.5) *Bangalore, India*